Upright kNN: Completing Nearest-Neighbour Density on Diffusion Features with the Missing Orientation Factor for Out-of-Distribution Detection

Anonymous Submission

Abstract

Diffusion models are attractive for unsupervised out-of-distribution (OOD) detection, but their likelihoods are fooled by image complexity and their score statistics cost tens to hundreds of network evaluations per image. A single denoiser pass already gives strong features: a k-nearest-neighbour (kNN) rule on channel statistics of decoder activations beats such statistics, yet these features barely encode which way is up, so an upside-down in-distribution image still looks in-distribution. We show that the log-density of an image equals, exactly, the log of its rotation-averaged density plus the log-posterior that the image is upright. Upright kNN estimates the first term with a kNN density on diffusion features and the second with a small MLP trained on rotated in-distribution images in the same feature space, and adds them with weights fixed by the decomposition. Under a protocol frozen before a single evaluation on held-out images (six in-distribution datasets, 63 pairs, 11 prior methods), Upright kNN reaches 93.4% mean AUROC against 85.5% for the best prior method, with one network evaluation per image. On the OpenOOD v1.5 ImageNet-1K benchmark it raises near-/far-OOD AUROC from 60.7/75.0% to 70.2/79.0% at 64×64; at 256×256, with a different diffusion model and no retuning, it is best on near-OOD NINCO and on upside-down images and ties the best prior method on far-OOD. Naive fusions of kNN with a rotation classifier fall short of it.

1 Introduction

A generative model of the training data offers a label-free test for inputs that do not come from the training distribution (out-of-distribution, OOD): such inputs should have low likelihood. Deep generative models, however, often assign higher likelihood to OOD images than to in-distribution (ID) ones [Nalisnick *et al.*, 2019a], largely because likelihood tracks image complexity [Serrà *et al.*, 2020]. Diffusion models [Ho *et al.*, 2020; Song *et al.*, 2021] inherit the problem. Detectors built on their score fields, sampling paths or posterior statistics [Mahmood *et al.*, 2021; Heng *et al.*, 2024; Rouzoumka *et al.*, 2026b; Shoushtari *et al.*, 2026] need 10 to 256 network evaluations per image and remain unstable across ID datasets: in our evaluation, the best of eight such detectors averages 80.5% AUROC over six ID datasets (Table 1).

A single denoiser pass is a cheaper and stronger source of evidence. Intermediate activations of a diffusion denoiser are good image representations [Baranchuk *et al.*, 2022; Xiang *et al.*, 2023], and CFS [Rouzoumka *et al.*, 2026a] showed that channel statistics of a few blocks, scored against ID data, beat likelihood- and statistic-based detectors. The kNN rule of Sun *et al.* [2022] applied to such features is stronger still (92.6% with a tuned configuration). These features have a blind spot, though: they barely encode which way is up. The tuned kNN separates vertically flipped ID images—same content, wrong orientation—from the originals with only 50–77% AUROC on four of the six ID datasets. For natural images, being upright is part of what makes an image typical, i.e. part of the density $p(\mathbf{x})$.

We make this intuition exact. Let $\bar{p}$ be the density averaged over the four rotations by multiples of 90°. Under the mixture of rotated ID images, the posterior probability that $\mathbf{x}$ is upright is $P(\mathrm{up} \mid \mathbf{x}) = p(\mathbf{x})/\{4\bar{p}(\mathbf{x})\}$, so that $\log p = \log \bar{p} + \log P(\mathrm{up} \mid \cdot) + \log 4$ holds for every image (Proposition 1). Channel statistics hardly change under rotation, so a kNN density on them mainly estimates the first, orientation-free term; the second term is what is missing. **Upright kNN** estimates it with a small MLP that is trained, on the features of the same single pass, to predict the rotation of rotated ID images, and it adds the two log-terms with weights fixed by the decomposition (Figure 1).

We evaluate under a protocol designed against selection bias: every configuration choice was made on separate selection data, the protocol was frozen, and the final images were evaluated once. Over six ID datasets with public diffusion models (63 ID–OOD pairs), Upright kNN reaches 93.4% mean AUROC, 7.9 points above the best of 11 prior methods, and it wins 41–62 of the 63 pairs against each of them at the cost of one network evaluation per image. On ImageNet-1K it improves near- and far-OOD AUROC over the best prior method at 64×64, and at 256×256, with another diffusion model, it is best on near-OOD NINCO and the only method that detects upside-down images. Our contributions are:

- An exact orientation decomposition of the log-density

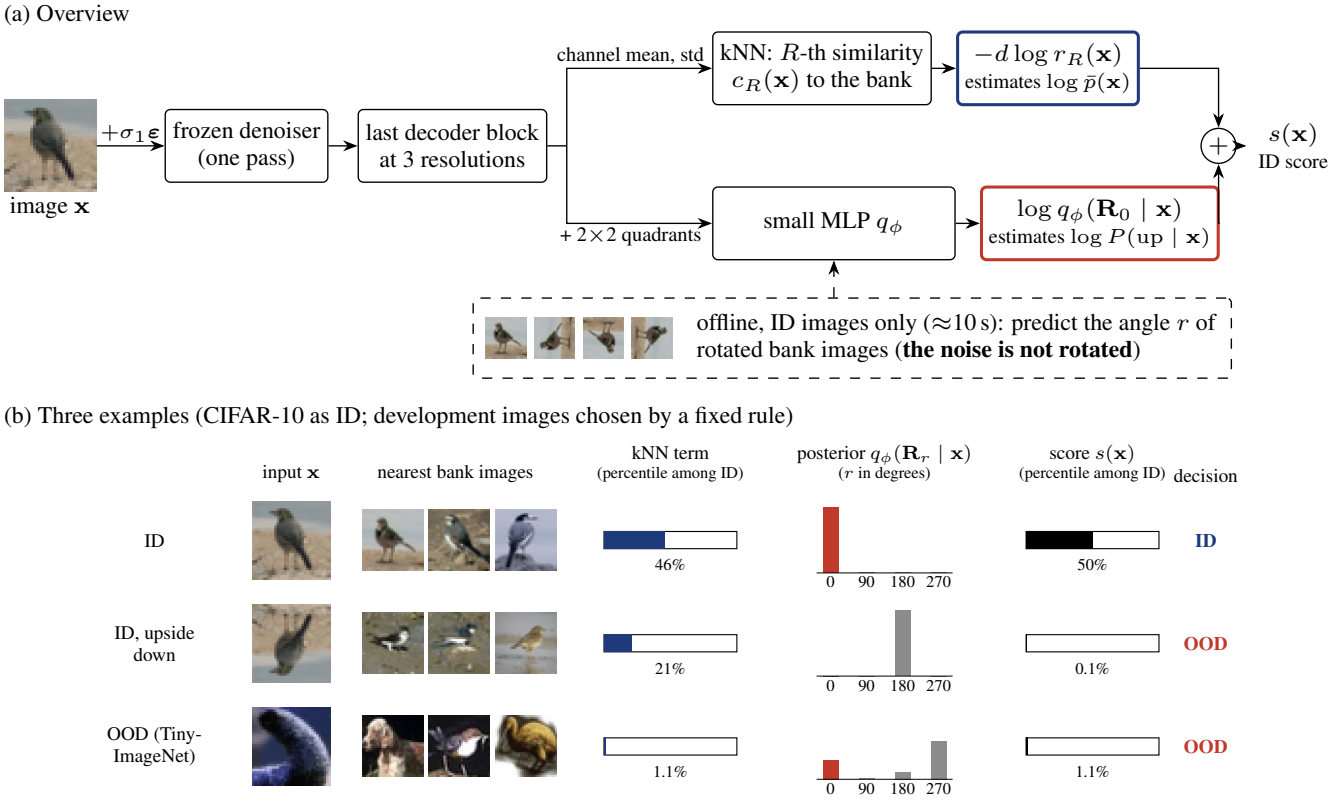


Figure 1: Upright kNN. (a) At test time, one denoiser pass gives the features of both terms of the score, Eq. (5). (b) From top: the ID image with the median score, the same image upside down, and the Tiny-ImageNet image with the median score of its set. Percentile: fraction of ID images with a lower value; decision: threshold that keeps 95% of ID images. The flipped image still has a bird as its nearest bank image and its kNN term stays at the 21st percentile, but its orientation posterior moves to 180° and its score drops to the 0.1st percentile.

and Upright kNN, a single-pass, label-free detector that restores the missing orientation factor to a nearest-neighbour density on diffusion features without tuned weights (Section 3).

- A frozen-protocol evaluation on six ID datasets against 11 prior methods, and on OpenOOD v1.5 ImageNet-1K with near-, far-, covariate-shift and full-spectrum metrics at 64×64 and, with a different diffusion model, at 256×256 (Sections 4–5).
- An analysis of what the orientation factor adds: naive fusions of kNN with a rotation classifier fall short, the exchange rate fixed by the decomposition matters, upside-down images are caught by the second term alone (Proposition 2), and a label-free mirror test locates orientation information inside the network (Section 6).

2 Related Work

Likelihoods and score statistics. Responses to the likelihood paradox [Nalisnick *et al.*, 2019a] include likelihood ratios [Ren *et al.*, 2019], typicality tests [Nalisnick *et al.*, 2019b], complexity corrections [Serrà *et al.*, 2020] and local intrinsic dimension [Kamkari *et al.*, 2024]. Diffusion-specific detectors use multiscale score norms (MSMA) [Mahmood *et al.*, 2021], reconstruction or inpainting errors [Graham *et al.*, 2023; Liu *et al.*, 2023], statistics of the sampling path (DiffPath) [Heng *et al.*, 2024], consistency of the posterior mean under transformations (GEPC) [Rouzoumka *et al.*, 2026b] and eigenvalues of the posterior covariance (EigenScore) [Shoushtari *et al.*, 2026]. Most need 10 to hundreds of network evaluations per image.

Feature-based detection. With supervised classifiers, Mahalanobis [Lee *et al.*, 2018] and deep kNN [Sun *et al.*, 2022] scores on penultimate features are strong baselines [Yang *et al.*, 2024; Zhang *et al.*, 2023]. Diffusion denoisers provide features without labels [Baranchuk *et al.*, 2022; Xiang *et al.*, 2023; Tang *et al.*, 2023]; CFS [Rouzoumka *et al.*, 2026a] pools channel statistics of one encoder and one decoder block at a single noise level and scores them with a diagonal Gaussian (or, in its appendix, a kNN head). Our kNN term uses the same kind of features; what we add is the orientation factor that the density decomposition shows to be missing.

Predicting transformations. Classifiers that predict rotations or other geometric transformations of ID images [Gidaris *et al.*, 2018] are used as anomaly scores [Golan and El-Yaniv, 2018; Hendrycks *et al.*, 2019], and contrastive methods such as CSI [Tack *et al.*, 2020] and SSD [Sehwag *et al.*, 2021] learn representations on ID data for the same purpose. These methods train a network from scratch and use the rotation cue as a stand-alone score or as a representation. We train only a small head on frozen diffusion features and use

its output as one factor of a density; Section 5.3 shows that combining kNN with a rotation classifier is not a substitute.

3 Method

Setting. We are given ID training images and a diffusion model trained on them; we use its denoiser $D(\mathbf{y};\sigma)$, no labels and no OOD data. A detector assigns a score $s(\mathbf{x})$ that is higher for ID-like images.

3.1 An Exact Orientation Decomposition

Let $\mathcal{G} = \{\mathbf{R}_0, \mathbf{R}_{90}, \mathbf{R}_{180}, \mathbf{R}_{270}\}$ be the rotations of the image plane by multiples of 90° and define the rotation-averaged density

$$\bar{p}(\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{G}\mathbf{x}). \tag{1}$$

Consider the mixture obtained by rotating ID images by an angle drawn uniformly from $\mathcal{G}$. Rotations permute pixels, so they preserve volume, and $\mathcal{G}$ is closed under inversion; hence the mixture has density $\frac{1}{4}\sum_{\mathbf{G}} p(\mathbf{G}^{-1}\mathbf{x}) = \bar{p}(\mathbf{x})$, and by Bayes' rule the posterior probability that $\mathbf{x}$ was not rotated is

$$P(\mathrm{up} \mid \mathbf{x}) = \frac{p(\mathbf{x})}{4\,\bar{p}(\mathbf{x})}. \tag{2}$$

Proposition 1. *For every image* $\mathbf{x}$ *with* $p(\mathbf{x}) > 0$*,*

$$\log p(\mathbf{x}) = \log \bar{p}(\mathbf{x}) + \log P(\mathrm{up} \mid \mathbf{x}) + \log 4. \tag{3}$$

The identity follows from Eq. (2) by taking logarithms; it is exact and involves no approximation. The first term is a *content* density that does not depend on orientation, since $\bar{p}(\mathbf{G}\mathbf{x}) = \bar{p}(\mathbf{x})$ for every $\mathbf{G} \in \mathcal{G}$; the second states how plausible the present orientation is, given the content. A detector that sees only rotation-invariant evidence can at best estimate the first term.

3.2 The Content Term: a kNN Density on Diffusion Features

We add noise $\sigma_1\boldsymbol{\varepsilon}$ to $\mathbf{x}$, with σ_1 the smallest noise level of the model (0.01 for all models here) and one noise image $\boldsymbol{\varepsilon}$ shared by all images, and pass $\mathbf{x} + \sigma_1\boldsymbol{\varepsilon}$ through the denoiser once. From the last decoder block at full, half and quarter resolution we take the channel-wise spatial mean and standard deviation and normalise each block's vector to unit length. The same features are computed for a bank of ID training images $\mathbf{b}_1, \ldots, \mathbf{b}_N$. Let $c_R(\mathbf{x})$ be the R-th largest similarity between $\mathbf{x}$ and the bank images (cosine similarity averaged over the three blocks) and $r_R(\mathbf{x}) = \sqrt{2\{1 - c_R(\mathbf{x})\}}$ the corresponding chordal distance. The kNN density estimate [Loftsgaarden and Quesenberry, 1965] in a space of dimension d is proportional to $r_R(\mathbf{x})^{-d}$, so

$$\log \hat{\bar{p}}(\mathbf{x}) = -d \log r_R(\mathbf{x}) + \mathrm{const}. \tag{4}$$

We set d to the maximum-likelihood intrinsic dimension [Levina and Bickel, 2004] of the bank features (20 neighbours; $d = 14 - -25$ across our ID datasets). Channel statistics are nearly invariant to rotation, so Eq. (4) estimates mainly the content term $\log \bar{p}$; Section 6 measures how much orientation information it still carries. On its own, Eq. (4) ranks images exactly as the kNN rule with similarity c_R does.

3.3 The Orientation Term: an Upright Posterior

We estimate the second term with a small MLP $q_\phi(\mathbf{R}_r \mid \mathbf{x})$ (one hidden layer of 512 units, four outputs) on features of the same pass: the channel means and standard deviations of the three blocks over the whole image and over each cell of a 2×2 grid, standardised on the bank. The grid statistics keep a coarse layout and thus an orientation cue. We hold out 5,000 bank images, rotate the remaining ones by each $\mathbf{R}_r \in \mathcal{G}$, and minimise the cross-entropy of predicting r. Because the four angles are equally likely, the minimiser is the posterior of the rotation mixture given the features; it equals Eq. (2) when the features keep the orientation information of the image.

Two details matter. First, the *noise is not rotated*: the inputs are $\mathbf{R}_r\mathbf{b}_i + \sigma_1\boldsymbol{\varepsilon}$. Since $\boldsymbol{\varepsilon}$ is shared, rotating it with the image would let the MLP read the angle from the noise pattern instead of the content; on ImageNet-64 this raises the held-out accuracy from 77.3% to 91.6% but lowers the AUROC on upside-down ID images from 82.6% to 71.9% (development images, before calibration). Second, we calibrate the posterior with a temperature T that minimises the negative log-likelihood on the held-out bank images [Guo *et al.*, 2017]. The head inputs and architecture were chosen by held-out rotation accuracy and likelihood only; the held-out accuracy is 77–100% across ID datasets.

3.4 The Score

Replacing both terms of Eq. (3) by their estimates gives

$$s(\mathbf{x}) = -d \log r_R(\mathbf{x}) + \log q_\phi(\mathbf{R}_0 \mid \mathbf{x}), \tag{5}$$

where constants are dropped because they do not affect rankings. No weight is tuned: the exchange rate between the terms is set by the intrinsic dimension d. Since the two terms are estimated on different summaries of the features and the kNN density lives in feature space, Eq. (5) is a plug-in combination guided by Eq. (3), not an unbiased estimate of $\log p$. At test time it costs one denoiser pass, inner products with the bank and a 2-ms head on a CPU; offline, it needs one pass over the four rotations of the bank and about 10 s of MLP training.

4 Experimental Setup

ID datasets and models. We use CIFAR-10 [Krizhevsky, 2009] (EDM [Karras *et al.*, 2022]), SVHN [Netzer *et al.*, 2011] (the VP-SDE of Kamkari *et al.* [2024]), CelebA [Liu *et al.*, 2015] (iDDPM [Nichol and Dhariwal, 2021]), CIFAR-100 (the EDM-preconditioned DDPM++ of Chen *et al.* [2024]), ImageNet-64 [Chrabaszcz *et al.*, 2017] (EDM2-XS, unconditional [Karras *et al.*, 2024]) and FashionMNIST [Xiao *et al.*, 2017] (a U-Net we trained). All other weights are public and used unchanged. All images are 32×32 except ImageNet-64.

OOD sets. Each ID dataset is paired with natural OOD datasets (9, 9, 9, 9, 7 and 4 sets; e.g. SSB-hard, NINCO, iNaturalist, Textures, OpenImage-O, Places365 and CelebA for ImageNet-64, and greyscale digit and character sets for FashionMNIST), with its own images flipped upside down, and with uniform-noise and constant images (except ImageNet-64): 63 ID–OOD pairs in total.

Protocol. Images are split into three disjoint parts: (i) selection data (ID development images and 6–11 OOD datasets per ID dataset that are never used for evaluation), (ii) development evaluation images (1,000 per set) and (iii) final evaluation images (500 per set). All configuration choices of Upright kNN (R and the blocks of the kNN term) were made on (i), the method was developed on (ii), and the written protocol was frozen before (iii) was evaluated once; all results below are on (iii) unless stated otherwise. The bank is 50,000 ID training images (10,000 for ImageNet-64, 10 per class).

Comparison methods. With the same diffusion models we run the likelihood (NLL, by the probability-flow ODE), input complexity with PNG and JPEG2000 codes (IC) [Serrà *et al.*, 2020], the typicality test (TT) [Nalisnick *et al.*, 2019b], a Mahalanobis version of MSMA (MSMA*) [Mahmood *et al.*, 2021], DiffPath [Heng *et al.*, 2024], GEPC [Rouzoumka *et al.*, 2026b], EigenScore [Shoushtari *et al.*, 2026] and CFS with its diagonal and kNN heads [Rouzoumka *et al.*, 2026a], ported from the official code and run on our bank with the noise level of the official configuration (it is better than the paper's default on (ii)). We also train a ResNet-18 rotation classifier from scratch on the ID training images [Hendrycks *et al.*, 2019]. Finally, we apply the kNN rule of Sun *et al.* [2022] to our features: a basic version (half-resolution block, 10th neighbour) and a tuned version whose block, noise level, neighbour statistic and R are chosen per ID dataset on (i) among 80 configurations.

Metrics. We report AUROC (ID as positive) and FPR at 95% TPR. Upright kNN, both kNN rules, CFS, GEPC and EigenScore are averaged over three noise seeds. For pairwise comparisons we bootstrap ID and OOD images (2,000 replicates) and count a win (loss) when the 95% interval of the AUROC difference excludes zero and the difference is at least 0.05 points.

5 Results

5.1 Six ID Datasets

Table 1 gives the final evaluation. Upright kNN averages 93.4% AUROC over the six ID datasets, 7.9 points above the best prior method (CFS with a kNN head, 85.5%); the rotation classifier reaches 84.0% and the best likelihood or statistic-based detector 80.5% (MSMA*). The kNN rule on our features is a strong baseline in its own right (87.7% basic, 92.6% tuned), and Upright kNN improves on the tuned version most where orientation matters, CIFAR-10 (95.4 vs 93.7%) and ImageNet-64 (79.6 vs 77.0%). Against each prior method it wins 41–62 of the 63 pairs and loses at most 4; against the tuned kNN it scores 30/32/1. It loses to no comparison method on 56 pairs. The 7 lost pairs are four upside-down sets (all to the rotation classifier), two pairs won by likelihood-based detectors (SVHN vs CelebA; ImageNet-64 vs iNaturalist, 84.4% NLL vs 71.9% ours) and one pair won by the tuned kNN (CIFAR-100 vs CIFAR-10). The mean FPR95 is 26.5% for Upright kNN, 29.8% for the tuned kNN and 43.5% for the best prior method (Rotation pred.).

On upside-down ID images (Figure 2), Upright kNN averages 86.6% against 72.7% for the tuned kNN and 70.6% for CFS (kNN head), e.g. 80.4% vs 50.5% on ImageNet-64. The rotation classifier is higher (94.6%), and so is the orientation term alone (94.7%): in the sum, the kNN term varies from image to image while staying almost unchanged under flipping, which dilutes the drop of the orientation term. Upright kNN takes 0.18 s per image on a CPU, about 1/320 of the likelihood (58 s) and 1/11 of DiffPath (2.0 s).

5.2 ImageNet-1K (OpenOOD v1.5)

To test the method at scale we apply the frozen ImageNet-64 detectors, without any refitting, to the OpenOOD v1.5 ImageNet-1K benchmark [Zhang *et al.*, 2023]: ID is ImageNet validation [Russakovsky *et al.*, 2015]; near-OOD is SSB-hard [Vaze *et al.*, 2022] and NINCO [Bitterwolf *et al.*, 2023]; far-OOD is iNaturalist [Huang and Li, 2021], Textures [Cimpoi *et al.*, 2014] and OpenImage-O [Wang *et al.*, 2022]; and the covariate-shifted ID sets (csID) are ImageNet-V2 [Recht *et al.*, 2019], ImageNet-C [Hendrycks and Dietterich, 2019] and ImageNet-R [Hendrycks *et al.*, 2021]. Images used earlier for development or selection are excluded, leaving 42,000 ID images; Table 2 uses a fixed subsample of 1,000 images per set on which every method was run (on the full sets, Upright kNN gives 70.6% near and 79.3% far).

At 64×64, Upright kNN reaches 70.2% near-OOD and 79.0% far-OOD AUROC, against 60.7% and 75.0% for the best prior method (CFS). It wins at least 4 of the five OOD sets against every prior method and loses only on iNaturalist, to NLL and TT (83.8% and 74.0% vs 70.1%). Both terms contribute: the kNN term alone gives 69.5/76.4% and the orientation term alone 63.5/77.4%.

The csID columns show what a density-based detector does under covariate shift. ImageNet-V2 is treated as ID (52.4%), whereas corrupted and rendered images are flagged (ImageNet-C 92.3%, rising with severity from 88.4% to 94.5%; ImageNet-R 77.9%). The full-spectrum protocol [Yang *et al.*, 2023] counts these as errors, so there Upright kNN (53.9/63.3%) trails NLL (56.5/69.3%), which ranks corrupted and rendered images as *more* likely than clean ones (43.3% and 31.3%). Whether covariate shift should be flagged depends on the application; Upright kNN flags it.

256×256 with a different diffusion model. We repeat the benchmark with the unconditional ADM model for 256×256 ImageNet [Dhariwal and Nichol, 2021], keeping every choice of the ImageNet-64 configuration ($R = 1000$, the three blocks, σ_1, the head and its training) and a bank of 10 training images per class; all rows are evaluated on the 500 images per set for which the stored MSMA*, DiffPath and GEPC statistics exist (Table 3). SSB-hard and ImageNet-C are distributed at 224 pixels on the short side and must be upsampled, unlike ImageNet validation (median 375 pixels) and the other sets. Every feature-based detector then separates SSB-hard almost perfectly (93.7% for ours, 85.4% for CFS), a resolution shortcut rather than semantic novelty, so we leave both sets out at this resolution.[1] Upright kNN is best on NINCO (80.6% vs 65.9% for the best prior method) and on upside-down images (78.5%, while every other method stays at chance, at

[1] At 256×256, the 32-image probe that CFS uses to rank its candidate blocks is passed through the network in chunks of four to fit GPU memory; everything else follows the official code.

Method	ID dataset (mean AUROC over its natural OOD sets) CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip	W/T/L of ours	NFE	CPU s/img
NLL	58.2	98.9	79.9	53.8	52.3	33.3	62.7	56.6	54/7/2	128+128	58
IC (PNG)	73.4	92.1	87.3	66.5	47.1	86.2	75.4	57.1	56/7/0	128+128	58
IC (JPEG2000)	71.4	74.1	82.0	65.2	46.4	40.8	63.3	56.6	56/7/0	128+128	58
TT	56.8	98.6	85.1	51.4	58.1	61.6	68.6	53.9	55/7/1	128+128	58
MSMA*	63.6	99.1	99.7	61.3	63.2	96.4	80.5	63.3	51/12/0	20	4.3
DiffPath	71.0	99.5	99.9	64.9	54.9	80.2	78.4	62.3	43/20/0	10	2.0
GEPC	51.9	91.3	99.1	55.6	51.2	84.4	72.3	52.1	58/5/0	14	2.9
EigenScore	73.4	85.8	86.9	47.6	44.2	94.3	72.0	75.4	62/0/1	180	40
CFS (diag.)	73.8	90.7	99.7	65.3	67.2	97.6	82.4	59.3	50/13/0	1	0.21
CFS (kNN head)	84.9	92.1	99.9	72.3	63.9	99.9	85.5	70.6	48/15/0	1	0.24
Rotation pred.	85.3	95.4	**100.0**	61.8	62.1	99.4	84.0	**94.6**	41/18/4	4‡	0.03
kNN rule [Sun et al.*, 2022] on our diffusion features*											
kNN (basic)	88.6	98.6	**100.0**	80.3	59.1	99.7	87.7	68.2	45/18/0	1	0.18
kNN (tuned)	93.7	99.4	**100.0**	85.4	77.0	99.9	92.6	72.7	30/32/1	1	0.18
Upright kNN (ours)	**95.4**	**99.7**	**100.0**	**85.5**	**79.6**	**100.0**	**93.4**	86.6	–	1	0.18
kNN term only	93.7	99.4	100.0	85.0	77.0	99.9	92.5	72.8	34/28/1	1	–
orientation term only	93.8	95.7	100.0	72.9	77.4	99.8	89.9	94.7	32/22/9	1	–

Table 1: Final evaluation (AUROC, %; 500 images per set). Columns 2–7: mean over the natural OOD sets of each ID dataset; Mean: their average; Flip: average over the six upside-down ID sets. W/T/L: wins/ties/losses of Upright kNN against the row over all 63 pairs. NFE: network evaluations per image (forward+backward; ‡ResNet-18 passes). CPU: seconds per image with the CIFAR-10 network (2 threads). Best and second best per column (over all rows except the two single-term rows) are marked **blue** and underlined.

Method	OOD Near	OOD Far	csID V2	csID C	csID R	Full spectrum Near	Full spectrum Far
Upright kNN	**70.2**	**79.0**	52.4	92.3	77.9	53.9	63.3
kNN term	69.5	76.4	52.2	92.1	73.4	55.0	61.9
orient. term	63.5	77.4	50.9	80.7	77.8	50.0	64.8
kNN (basic)	60.5	68.1	50.6	83.6	63.9	51.9	60.4
CFS (diag.)	60.7	75.0	52.0	77.3	68.7	51.4	65.1
CFS (kNN)	56.3	71.0	51.5	80.7	67.0	46.8	63.0
Rotation pred.	53.1	70.3	49.7	63.2	65.1	47.1	64.4
NLL	50.4	64.9	51.0	43.3	31.3	**56.5**	**69.3**
IC (PNG)	52.8	57.1	48.1	56.6	45.7	53.5	57.5
IC (J2K)	51.2	55.7	48.3	59.9	36.1	54.7	58.4
TT	56.7	63.4	49.6	66.1	62.7	50.4	56.2
MSMA*	56.7	59.8	46.9	79.1	72.1	46.1	49.8
DiffPath	53.3	62.6	52.1	57.6	69.6	45.7	54.0
GEPC	50.0	56.4	49.8	57.8	59.0	46.3	52.8
EigenScore	46.4	46.0	49.5	54.0	67.8	40.2	40.2

Table 2: OpenOOD v1.5 ImageNet-1K at 64×64 with EDM2-XS (AUROC, %; 1,000 images per set; frozen ImageNet-64 detectors). Near: SSB-hard, NINCO; far: iNaturalist, Textures, OpenImage-O; csID: ImageNet-V2, -C, -R against ImageNet val (ID); full spectrum: ID and csID images together as the positive class, weighted by the benchmark's set sizes. Marks as in Table 1; the csID columns are not marked.

Method	Near NINCO	Far iNat	Far Tex	Far OI-O	Far Mean	Flip	W/T/L of ours
Upright kNN	**80.6**	84.1	78.5	54.0	72.2	**78.5**	–
kNN term	81.4	86.2	75.1	48.6	69.9	50.6	2/1/1
orient. term	66.1	61.3	77.2	71.5	70.0	92.6	2/1/1
kNN (basic)	66.1	**85.6**	50.3	13.0	49.7	50.3	3/1/0
CFS (diag.)	62.5	57.3	**88.5**	76.5	**74.1**	49.4	2/0/2
CFS (kNN)	65.9	57.9	82.1	41.6	60.5	50.7	3/0/1
MSMA*	62.0	33.9	88.2	**79.5**	67.2	50.0	2/0/2
DiffPath	53.7	39.3	60.6	44.9	48.3	50.8	4/0/0
GEPC	54.0	48.4	49.1	40.0	45.8	50.8	4/0/0

Table 3: ImageNet at 256×256 with ADM (AUROC, %; 500 common images per set; the ImageNet-64 configuration unchanged; 3 noise seeds for our rows and CFS). Flip: upside-down ID images. SSB-hard and ImageNet-C are left out because their 224-pixel images would be upsampled (see text). W/T/L over the four OOD sets; marks as in Table 1.

most 50.8%), and second on far-OOD (72.2% vs 74.1% for CFS, a tie under the bootstrap). Its weak spot is OpenImage-O (54.0% vs 76.5% for CFS), where the kNN term itself fails. At this resolution the kNN term carries most of the near-OOD signal (81.4% on NINCO alone) and the orientation term the flip signal (92.6%).

5.3 Is It Just kNN Plus a Rotation Classifier?

Upright kNN combines a nearest-neighbour score with a rotation predictor, so a natural question is whether any such combination works as well. Table 4 compares the sum of Eq. (5) with fusions that standardise the two scores using additional ID images. Fusing the kNN term with the image-space rotation classifier gives 91.2% (z-sum) and 91.2% (Fisher), *below* the kNN term alone; Upright kNN beats the z-sum fusion on 30/17/0 (W/T/L) of the natural-OOD pairs. The classifier's confidence is high on upside-down images (94.6%) but poorly aligned with semantic novelty, so it adds noise to the kNN term on natural OOD sets. Fusing our own two terms is much closer (93.1%), but it needs extra ID images for normalisation, still loses more pairs than it wins against the principled sum (23/17/7), and trades natural-OOD accuracy for flip detection. The exchange rate matters: setting $d = 1$ in Eq. (5) gives 92.0% (24/20/3). On the development images, further variants are also worse on average: no temperature (93.3%),

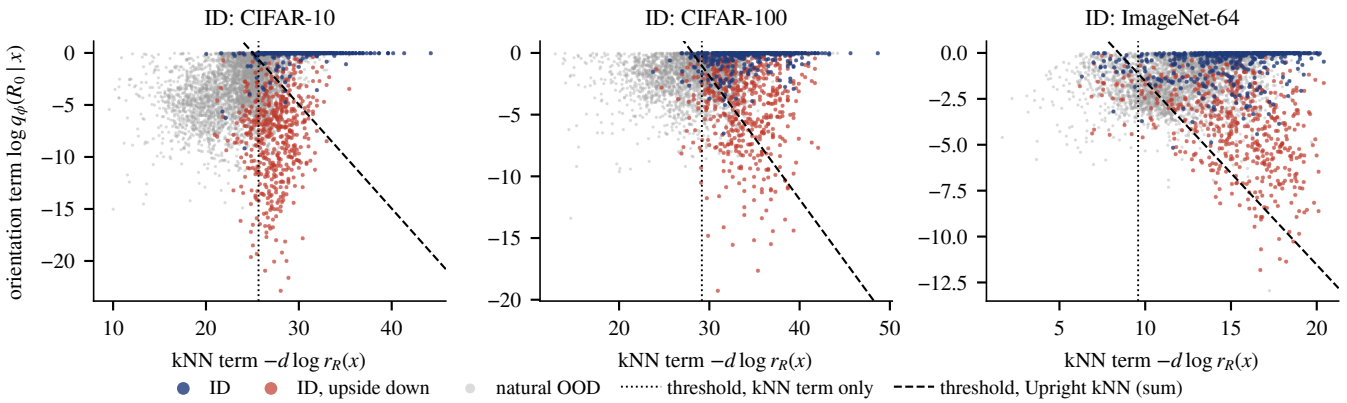


Figure 2: The two terms of Upright kNN on the final evaluation images (noise seed 0) for the three ID datasets with the lowest AUROC. Dotted: threshold of the kNN term alone that keeps 95% of the ID images; dashed: the same for their sum (slope -1); images below and left of a line are declared OOD. Upside-down ID images (red) overlap the ID images (navy) in the kNN term but fall below the dashed line through the orientation term, whereas natural OOD images (grey) are separated mainly by the kNN term.

Score	Mean	Flip	W/T/L of ours
Upright kNN (sum, Eq. 5)	**93.4**	86.6	–
kNN term only	92.5	72.8	28/18/1
orientation term only	89.9	94.7	29/14/4
rotation classifier (ResNet-18)	84.0	94.6	35/12/0
kNN + rot. classifier, z-sum	91.2	92.4	30/17/0
kNN + rot. classifier, Fisher	91.2	92.9	19/26/2
kNN + orient. term, z-sum	93.1	92.4	23/17/7
kNN + orient. term, Fisher	93.1	92.5	13/26/8
Upright kNN with $d = 1$	92.0	94.2	24/20/3

Table 4: Principled sum vs. naive fusions (final evaluation, AUROC, %; Mean and Flip as in Table 1). z-sum: sum of z-scores; Fisher: sum of log empirical ID p-values [Fisher, 1925]; both normalise each score with additional ID images (the development split), which Upright kNN does not need. W/T/L over the 47 natural-OOD pairs. This comparison was added after the frozen evaluation and reuses its images.

estimating $\bar{p}$ directly with a rotated bank (92.6%; 91.8% vs 95.9% on CIFAR-10), training the head with rotated noise (92.9%), or averaging the orientation term over four rotated test views (92.9%), against 93.4% for Upright kNN; none of them raises the selection-data mean either.

6 Analysis

Why upside-down images are caught. A vertical flip $\mathbf{V}$ equals a horizontal flip $\mathbf{H}$ composed with a half turn, $\mathbf{V} = \mathbf{H}\mathbf{R}_{180}$. Natural ID data are close to left–right symmetric, which gives the following.

Proposition 2. *If* $p(\mathbf{H}\mathbf{x}) = p(\mathbf{x})$ *for all* $\mathbf{x}$*, then* $\bar{p}(\mathbf{V}\mathbf{x}) = \bar{p}(\mathbf{x})$ *and* $\log p(\mathbf{V}\mathbf{x}) - \log p(\mathbf{x}) = \log P(\text{up} \mid \mathbf{V}\mathbf{x}) - \log P(\text{up} \mid \mathbf{x})$.

Proof. For $\mathbf{G} \in \mathcal{G}$ we have $\mathbf{G}\mathbf{H} = \mathbf{H}\mathbf{G}^{-1}$. Hence $\bar{p}(\mathbf{V}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}} p(\mathbf{H}\mathbf{G}^{-1}\mathbf{R}_{180}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}} p(\mathbf{G}^{-1}\mathbf{R}_{180}\mathbf{x}) = \bar{p}(\mathbf{x})$, because $\mathbf{G}^{-1}\mathbf{R}_{180}$ runs over $\mathcal{G}$ once as $\mathbf{G}$ does. The second claim follows by subtracting Eq. (3) at $\mathbf{x}$ from the same equation at $\mathbf{V}\mathbf{x}$. □

A flip therefore changes only the orientation term; a good content term is blind to it. This is what Figure 1(b) shows for one image (the flipped bird keeps its nearest neighbours, while the posterior moves to 180°) and Figure 2 for all of them: flipped ID images stay in the ID range of the kNN term and are pushed out by the orientation term. The proposition also explains the failure on SVHN (flip AUROC 64.5%; 78.6% for the orientation term alone): many digits (0, 1, 3, 8) look the same upside down.

A mirror test for orientation information. Proposition 2 suggests a label-free probe. For any score e and ID development images $\mathbf{x}_1, \ldots, \mathbf{x}_n$, let $M(e)$ be the AUROC between $\{e(\mathbf{x}_i)\}$ and $\{e(\mathbf{V}\mathbf{x}_i)\}$. An ideal content term has $M = 50\%$; M near 100% means that e prefers upright images. For the kNN term computed from single blocks ($R = 1$), M grows with depth (CIFAR-10: 52.1%, 67.3% and 87.8% at full, half and quarter resolution), while the same test with a horizontal flip stays at 49–56% on the natural-image ID datasets. The kNN term of Upright kNN therefore still carries some orientation information (on CIFAR-10, $M = 77.7\%$), which the orientation term complements. Across all blocks, M is rank-correlated with the block's natural-OOD AUROC ($\rho = 0.68$–0.91 on five ID datasets; -0.53 on SVHN, whose OOD sets differ mainly in colour and texture), so ID images alone indicate which blocks are useful for OOD detection.

Data without a canonical orientation. To see what happens when the ID data have no meaningful *up*, we ran Upright kNN on EuroSAT aerial tiles [Helber *et al.*, 2019] with a diffusion U-Net trained on them, ten natural OOD sets and $R = 5$ fixed in advance (a few duplicate tiles, whose neighbours all coincide, are skipped when estimating d). We expected the head to fall to chance; instead it predicts the rotation of held-out tiles with 72–74% accuracy (chance: 25%), presumably from illumination and north-up acquisition. The orientation term therefore flags rotated tiles (75.8% AUROC for 90° rotations, against 55.1% for the kNN term), while natural-OOD

detection is unchanged (99.8% for Upright kNN and 99.8% for its kNN term). Upright kNN thus estimates whatever orientation statistics the ID data have; when rotated inputs must count as ID, the orientation term should be dropped.

Where the orientation term hurts. The orientation term also reflects how familiar the content is to the ID model, which usually helps (CIFAR-10 vs CIFAR-100: 86.0% for the orientation term alone vs 77.4% for the kNN term) but can hurt: with CIFAR-100 as ID, the head finds CIFAR-10 images *more* upright than CIFAR-100 images (orientation-term AUROC 28.4%), which causes the single loss to the tuned kNN (64.1 vs 66.6%).

7 Limitations and Conclusion

The orientation term encodes whatever orientation statistics the ID data have: it adds little when the content looks alike upside down (SVHN digits), and on aerial tiles it flags rotated inputs, which is unwanted when rotations should count as ID. Near-OOD remains hard: on ImageNet-64 the near-OOD AUROC stays around 70%, and on the development images even the best of 2,738 scores we computed, selected per pair after the fact, stays below 80% on CIFAR-100 vs CIFAR-10 and on SSB-hard and NINCO. The detector flags covariate shift, which full-spectrum OOD detection penalises, and at high resolution benchmark sets stored at a lower resolution than the ID images give every feature-based detector a shortcut that comparisons must control for. Finally, the bank features must be stored and searched, at a cost linear in the bank size.

Within these limits, a simple observation goes a long way: the log-density splits exactly into a rotation-invariant content term and an orientation term, and the single-pass features of a diffusion denoiser estimate the first well and, with a small head, the second as well. Restoring the missing factor turns a strong kNN baseline into a detector that beats 11 prior methods on six ID datasets under a frozen protocol and carries over to ImageNet at 64×64 and, with another diffusion model, at 256×256, at the cost of one network evaluation per image.

References

[Baranchuk *et al.*, 2022] Dmitry Baranchuk, Andrey Voynov, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Label-efficient semantic segmentation with diffusion models. In *ICLR*, 2022.

[Bitterwolf *et al.*, 2023] Julian Bitterwolf, Maximilian Müller, and Matthias Hein. In or out? Fixing ImageNet out-of-distribution detection evaluation. In *ICML*, 2023.

[Chen *et al.*, 2024] Huanran Chen, Yinpeng Dong, Zhengyi Wang, Xiao Yang, Chengqi Duan, Hang Su, and Jun Zhu. Robust classification via a single diffusion model. In *ICML*, 2024.

[Chrabaszcz *et al.*, 2017] Patryk Chrabaszcz, Ilya Loshchilov, and Frank Hutter. A downsampled variant of ImageNet as an alternative to the CIFAR datasets. *arXiv preprint arXiv:1707.08819*, 2017.

[Cimpoi *et al.*, 2014] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *CVPR*, 2014.

[Dhariwal and Nichol, 2021] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *NeurIPS*, 2021.

[Fisher, 1925] Ronald A. Fisher. *Statistical Methods for Research Workers*. Oliver and Boyd, 1925.

[Gidaris *et al.*, 2018] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations. In *ICLR*, 2018.

[Golan and El-Yaniv, 2018] Izhak Golan and Ran El-Yaniv. Deep anomaly detection using geometric transformations. In *NeurIPS*, 2018.

[Graham *et al.*, 2023] Mark S. Graham, Walter H. L. Pinaya, Petru-Daniel Tudosiu, Parashkev Nachev, Sebastien Ourselin, and M. Jorge Cardoso. Denoising diffusion models for out-of-distribution detection. In *CVPR Workshops*, 2023.

[Guo *et al.*, 2017] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[Helber *et al.*, 2019] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.

[Hendrycks and Dietterich, 2019] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *ICLR*, 2019.

[Hendrycks *et al.*, 2019] Dan Hendrycks, Mantas Mazeika, Saurav Kadavath, and Dawn Song. Using self-supervised learning can improve model robustness and uncertainty. In *NeurIPS*, 2019.

[Hendrycks *et al.*, 2021] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, and Justin Gilmer. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *ICCV*, 2021.

[Heng *et al.*, 2024] Alvin Heng, Alexandre H. Thiery, and Harold Soh. Out-of-distribution detection with a single unconditional diffusion model. In *NeurIPS*, 2024.

[Ho *et al.*, 2020] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.

[Huang and Li, 2021] Rui Huang and Yixuan Li. MOS: Towards scaling out-of-distribution detection for large semantic space. In *CVPR*, 2021.

[Kamkari *et al.*, 2024] Hamidreza Kamkari, Brendan Leigh Ross, Jesse C. Cresswell, Anthony L. Caterini, Rahul G.

Krishnan, and Gabriel Loaiza-Ganem. A geometric explanation of the likelihood OOD detection paradox. In *ICML*, 2024.

[Karras *et al.*, 2022] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *NeurIPS*, 2022.

[Karras *et al.*, 2024] Tero Karras, Miika Aittala, Jaakko Lehtinen, Janne Hellsten, Timo Aila, and Samuli Laine. Analyzing and improving the training dynamics of diffusion models. In *CVPR*, 2024.

[Krizhevsky, 2009] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[Lee *et al.*, 2018] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *NeurIPS*, 2018.

[Levina and Bickel, 2004] Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *NeurIPS*, 2004.

[Liu *et al.*, 2015] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *ICCV*, 2015.

[Liu *et al.*, 2023] Zhenzhen Liu, Jin Peng Zhou, Yufan Wang, and Kilian Q. Weinberger. Unsupervised out-of-distribution detection with diffusion inpainting. In *ICML*, 2023.

[Loftsgaarden and Quesenberry, 1965] Don O. Loftsgaarden and Charles P. Quesenberry. A nonparametric estimate of a multivariate density function. *The Annals of Mathematical Statistics*, 36(3):1049–1051, 1965.

[Mahmood *et al.*, 2021] Ahsan Mahmood, Junier Oliva, and Martin Styner. Multiscale score matching for out-of-distribution detection. In *ICLR*, 2021.

[Nalisnick *et al.*, 2019a] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, Dilan Gorur, and Balaji Lakshminarayanan. Do deep generative models know what they don't know? In *ICLR*, 2019.

[Nalisnick *et al.*, 2019b] Eric Nalisnick, Akihiro Matsukawa, Yee Whye Teh, and Balaji Lakshminarayanan. Detecting out-of-distribution inputs to deep generative models using typicality. *arXiv preprint arXiv:1906.02994*, 2019.

[Netzer *et al.*, 2011] Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Bo Wu, and Andrew Y. Ng. Reading digits in natural images with unsupervised feature learning. In *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.

[Nichol and Dhariwal, 2021] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, 2021.

[Recht *et al.*, 2019] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *ICML*, 2019.

[Ren *et al.*, 2019] Jie Ren, Peter J. Liu, Emily Fertig, Jasper Snoek, Ryan Poplin, Mark A. DePristo, Joshua V. Dillon, and Balaji Lakshminarayanan. Likelihood ratios for out-of-distribution detection. In *NeurIPS*, 2019.

[Rouzoumka *et al.*, 2026a] Yadang Alexis Rouzoumka, Jean Pinsolle, Eugénie Terreaux, Christèle Morisseau, Jean-Philippe Ovarlez, and Chengfang Ren. Backbone-equated diffusion OOD via sparse internal snapshots. *arXiv preprint arXiv:2605.11014*, 2026.

[Rouzoumka *et al.*, 2026b] Yadang Alexis Rouzoumka, Jean Pinsolle, Eugénie Terreaux, Christèle Morisseau, Jean-Philippe Ovarlez, and Chengfang Ren. GEPC: Group-equivariant posterior consistency for out-of-distribution detection in diffusion models. In *ICML*, 2026.

[Russakovsky *et al.*, 2015] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.

[Sehwag *et al.*, 2021] Vikash Sehwag, Mung Chiang, and Prateek Mittal. SSD: A unified framework for self-supervised outlier detection. In *ICLR*, 2021.

[Serrà *et al.*, 2020] Joan Serrà, David Álvarez, Vicenç Gómez, Olga Slizovskaia, José F. Núñez, and Jordi Luque. Input complexity and out-of-distribution detection with likelihood-based generative models. In *ICLR*, 2020.

[Shoushtari *et al.*, 2026] Shirin Shoushtari, Yi Wang, Xiao Shi, M. Salman Asif, and Ulugbek S. Kamilov. EigenScore: OOD detection using posterior covariance in diffusion models. In *ICLR*, 2026.

[Song *et al.*, 2021] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, 2021.

[Sun *et al.*, 2022] Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *ICML*, 2022.

[Tack *et al.*, 2020] Jihoon Tack, Sangwoo Mo, Jongheon Jeong, and Jinwoo Shin. CSI: Novelty detection via contrastive learning on distributionally shifted instances. In *NeurIPS*, 2020.

[Tang *et al.*, 2023] Luming Tang, Menglin Jia, Qianqian Wang, Cheng Perng Phoo, and Bharath Hariharan. Emergent correspondence from image diffusion. In *NeurIPS*, 2023.

[Vaze *et al.*, 2022] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need. In *ICLR*, 2022.

[Wang *et al.*, 2022] Haoqi Wang, Zhizhong Li, Litong Feng, and Wayne Zhang. ViM: Out-of-distribution with virtual-logit matching. In *CVPR*, 2022.

[Xiang *et al.*, 2023] Weilai Xiang, Hongyu Yang, Di Huang, and Yunhong Wang. Denoising diffusion autoencoders are unified self-supervised learners. In *ICCV*, 2023.

[Xiao *et al.*, 2017] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

[Yang *et al.*, 2023] Jingkang Yang, Kaiyang Zhou, and Ziwei Liu. Full-spectrum out-of-distribution detection. *International Journal of Computer Vision*, 2023.

[Yang *et al.*, 2024] Jingkang Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 2024.

[Zhang *et al.*, 2023] Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yueqian Lin, Haoran Zhang, Yiyou Sun, Xuefeng Du, Yixuan Li, Ziwei Liu, Yiran Chen, and Hai Li. OpenOOD v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*, 2023.